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Use-Wear Analysis Experiment

To say that the history of use-wear analysis has been tumultuous would be an understatement. Since its inception in 1957, there have been many ups and downs in the field. Things seemed to be going in the right direction, with more people studying the topic into the 1960s and the first conferences being held in the 1970s. However, beginning in the 1980s, the study of use-wear analysis would start to face increasing scrutiny. Blind tests showed that the application of the technique and its results were not necessarily always clearly distinguishable by the 1990s, the use-wear analysis had experience quite the battering, and at this point the damage to its reputation was done. Nonetheless, there are many people who still utilize the technique today and find it quite valuable. But why is this?

Use-wear analysis can be effective with either a small assemblage, clearly distinguishable traits, or when used in tandem with residue analysis. In this experiment, use-wear analysis was conducted on both obsidian and red windblown sandstone, with a chef's knife as a control, in order to verify my hypothesis. These two materials were chosen due to the drastically different nature of their makeup and hardness. Ultimately, the hypothesis was confirmed, and use-wear analysis is most effective when applied specifically to a research question rather than used broadly as a catch-all technique.

Changes in the Field

Quite a lot has changed since Sergei Semenov's *Prehistoric Technology* was first released in 1957. While not deemed use-wear analysis at the time, this was the first work "derived from the analysis of linear abrasion, called striations, on modern metal implements" (Odell 2004 pg.136). His work would go on to inspire people in the 1960s to begin applying the analysis method to an increasing number of assemblages. By applying this method to lithic tools, Semenov paved the way for a new field of archaeology.

Interest in functional issues increased following the period of Semenov's influence. In the West researchers such as Rosenfeld, Brink, and Sonnenfeld focused more on specific topics as opposed to being comprehensive in their studies (Odell 2004). One of the first series of comprehensive experiments was conducted in 1966 by Keller. During the 1970s and 80s comprehensive experimental studies were conducted via dissecting microscopes, a technique which would eventually be called low-power analysis (Odell 2004). The first conference on lithic use-wear was held in Vancouver, British Columbia in 1977. This was the first time that scholars in this field got together to discuss their work and the future of this new field. The next method for use-wear analysis was developed by Keely using an "incident-light metallurgical microscope", a technique dubbed high-power analysis (Odell 2004).

Both high-power and low-power analysis would face challenges in their development. Despite John Tomenchuk's 1985 research employing engineering principles in his low-power analysis to provide a truly objective observation, low-power analysis became the "victim of misconception and misinformation" (Odell 2004). High-power analysis faced a number of issues as well. The formation and recognition of post-depositional surface modification (PDSM) or the alteration of tool surfaces resulting from factors such as soil chemistry, artifact cleaning

protocols, etc., which would have affected the tool since its discard, proved to be quite the blow to use-wear analysis in general. Many were skeptical that one could even distinguish between use-wear and PDSM. An experiment conducted in the 1980s by a team of multi-national use-wear specialists could not reproduce the results from two blind tests. Multiple issues along the way caused the community to wake up and make better attempts at use-wear analysis, employing all methods available. However, it was too late, and many were turned off to the field by the 1990s.

Use-wear analysis would recover by the late 1990s as the increased scrutiny it was put under caused the field to become quite thorough and well structured. Room for improvement remains today, but thankfully there are people willing to do the work. Issues that remain unsolved include "the effects of time and use independent of kinetics and worked material as well as the fact that use-wear found on retouched tools records that last use only, which may not be representative of earlier uses" (Shott pg.3).

Methods

Throughout the experiment, a variety of methods were implemented to properly test the hypothesis. Initial steps involved creating a chopper with both the obsidian and the wind-blown red sandstone. For the purposes of the experiment rosemary, thyme, a corn cob, and a T-bone steak were used as the items to process. A chef's knife was used as a control because use-wear analysis is based upon the principles used to examine use-wear on steel and other metal tools (Odell 2004). The only factors controlled for were the materials, the items processed, and the cutting board. Additionally, the obsidian tool was denticulated in order to study if this would create a noticeably different use-wear pattern than on the non-denticulated chopper. Other than that, everything was left to be variable. This was done to test if the hypothesis of use-wear being

best used with residue analysis unless the wear is obvious was correct. Had each object been used to process only one material, it would have been much easier to identify the use.

For the analysis, low-power and macroscopic analyses were employed. Additionally, the chef's knife was sharpened to compare the before and aftereffects of use more clearly, as the knife had seen prior use. Low-power analysis was done at both 10x magnification and 45x magnification in order to get both a general and close-up view of the tool edges. An i-Phone X was used to take photographs of the tools as well as the videos of the test that were conducted that will be included with this report. The microscope and space to conduct the experiment were both provided through the UNLV Archaeology lab via Professor Barbara Roth.

Results



Figure 1.1 – *dorsal side*



Figure 1.2 – *ventral side*



Figure 2 – *protective tools, knife, sharpener*



Figure 3 *– rosemary and thyme*



Figure 4 – *ears of corn*



Figure 5 – *steak*

Table 1 – pre-experiment

Pre-experiment observations	**Obsidian**	**Sandstone**	**Chef's Knife** (*control)*
	Clearly defined edges, some pieces appear loose	Clearly defined edges, not as smooth due to nature of sandstone, some pieces on edge appear loose	Initially sharpened to use as control Exhibited chipping on entire blade concentrated around the front and rear

			with excessive vertical and horizontal striations, sheen along blade edge After sharpening – noticeably reduced chipping, shavings visible at 10x mag., new striations from sharpening

Table 2 – *experimental results*

Post-experiment observations	**Obsidian**	**Sandstone**	**Chef's Knife** (*control*)
Rosemary & Thyme	Minimal striations, small hinge removals 45x mag. – major point of use has multiple hinge removals, visibly green	Minimal striations visible 45x mag. – some hinge removals, no sheening	Sheen along major point of contact 45x mag. – two significant hinge removals, sheen follows direction of use
Corn	Sheening visible at major point of contact, small hinge removals 45x mag. – consistent hinge removals visible around major point of contact, sheening follows the direction of use	Minimal chipping, covered in corn material 45x mag. – multiple hinge removals visible along major point of contact	Chipping, additional striations 45x mag. – Striations more visible along major point of contact, significant hinge removals
T-bone Steak	Striations visible with some thin termination points	Considerable chipping at major point of contact	Tip of knife was blunted, uniform reduction of blade

	45x mag. – striations and feather termination scarring much more visible, material on almost entire tool edge	45x mag. – some striations and significant feather termination scarring is visible as well as some hinge scarring	edge with some chipping 45x mag. – feather termination scarring starts from the middle of the blade, consistent with angle of use, hinge removals near tip of blade that was used for scraping bone

Videos: Click Here

Discussion

All three implements exhibited a variety of use-wear traits. However, the degree of expression between these traits varied greatly across the three tools used. For example, the obsidian displayed the clearest evidence of sheening likely due to it being a volcanic glass, something that tends to sheen more easily (Table 2). This is further supported by the fact that the windblown red sandstone displayed no sheening, likely due to the nature of sandstone being a sedimentary rock (Table 2). Upon cutting the corn, the windblown red sandstone was also the most resistant to showing signs of use-wear. Interestingly when it came to cutting the T-bone, the obsidian held up better than the sandstone. Despite the sandstone cutting through the steak with less resistance, it showed considerably more wear after cutting it than with the obsidian (Table 2).

While there were differences in the expression of use-wear between the three tools used, there were also some similarities that were observed. Upon cutting the rosemary and thyme, both the obsidian and the windblown red sandstone displayed striations along the major point of

contact. All three implements used to cut the rosemary and thyme also exhibited some form of hinge scarring, typical of a tool used to process plant material that is a bit harder to get through than just a leaf. When cutting the corn, the frequency, depth, and size of the hinge scarring on all tools increased, again a feature you would expect to see when processing tougher plant materials (Odell & Odell 1980). The consistency of feather termination scarring along major point of use for all tools is suggestive of use on meat with bone. The frequency of general scarring and striations would be significantly less across the board had the T-bone steak did not contain a bone.

Interestingly, all tools used in this experiment exhibited a great deal of material along their use-edges, especially when used with the meat. This suggests that including residue analysis along with use-wear analysis could be vital to determining the use of these tools, especially because they processed multiple kinds of material. If the plant and animal matter remained and was below 50,000 years old, the material could also be radiocarbon dated. Quite a lot of corn material was left on the tools as well, meaning residue analysis would again be useful here because you could determine what kind of corn it was.

When it comes to meat, there is quite a bit known about doing residue analysis on lithic tools. It has been shown that while all animals obviously carry blood, the shape and structure of the hemoglobin crystals left behind by blood residue is quite distinct based on the genus and sometime species of the animal (Shott). This information would be vital when conducting use-wear analysis on a tool suspected of being used in animal processing. If a researcher was uncertain about this, they could potentially use remaining residue on the tool to not only narrow down that the tool was used to process animals, but on exactly what kind of animal it was used.

The limited degree of control in this experiment was used to highlight issues related to the last use of the tool. When discovered, lithic artifacts are often found in the state of their last use. This creates difficulties when conducting use-wear analysis, as the final use of a tool may not be what it was originally intended for. A biface could initially be used to skin a hide, but it could then be reduced to a chopper or even a dart or arrowhead if the need arises. The only way you would be able to definitively verify the use of the artifact at this point would be to conduct a residue analysis. Had each tool been limited to only one type of material to be used on, it would have been much easier to analyze the use-wear. Because this is not the state that most tools are found in, the experiment repeated this vagueness of use to point out the issues inherent to this type of analysis.

Conclusion

Use-wear analysis has had a difficult but fruitful history. The early difficulties faced required the field to refine its guidelines and techniques more quickly than other disciplines, contributing to the effectiveness of its application. However, there are still issues that have yet to be solved. Post-depositional processes and last-use pose a serious problem for the continuing validity of the use-wear analysis methods currently available in the field of lithic analysis today. With further study into these issues and the application of residue analysis where use is unclear, use-wear analysis will continue to prove beneficial to employ in the coming years.

Works Cited

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